

Simulation results

massive-binary-sim v0.1.0 · git `UNKNOWN(no` · seed 42 · wallclock 0.07 s

1. Initial conditions

quantity	value
M1, M2 (ZAMS)	42, 33 Msun
separation / period	90 Rsun / 11.42 d
eccentricity	0
metallicity Z	0.003
spins ($\Omega/\Omega_{\text{crit}}$)	0.2, 0.2

Active prescriptions

toggle	choice
gravity	`2.5PN`
wind	`vink2001`
convection	`ledoux`
overshoot	`exponential`
am_transport	`taylor_spruit`
mt_stability	`zeta`
mt_rate	`kolb_ritter`
ce_formalism	`alpha_lambda`
opacity	`kramers`
reaction_rates	`reaclib`
sn_engine	`delayed`
tides	`hut1981`
structure_fidelity	`semi_analytic`
third_body	`False`

2. Evolutionary timeline

t [Myr]	event	detail
0.011	star 1 -> MS	reduced-model phase
4.237	star 1 -> HG	reduced-model phase
4.321	star 1 -> CHeB	reduced-model phase
4.747	star 1 -> preSN	reduced-model phase
4.757	star 1 -> remnant	reduced-model phase
0.011	star 2 -> MS	reduced-model phase
5.162	star 2 -> HG	reduced-model phase
5.256	star 2 -> CHeB	reduced-model phase
5.257	star 2 -> stripped_He	reduced-model phase
5.583	star 2 -> preSN	reduced-model phase
5.583	star 2 -> remnant	reduced-model phase
(at SN)	**SN1**	star 1 -> BH 36.78 Msun (delayed); bound after SN
5.256	**common_envelope**	survived: $a_f=24.654$ Rsun (spiral-in x4.0)
(at SN)	**SN2**	star 2 -> BH 6.71 Msun (delayed); bound after SN
5.583	**dco_formed**	double compact binary (BH+BH) merging via GW in 2216.3 Myr

Fate: double compact binary (BH+BH) merging via GW in 2216.3 Myr

3. Final-state table

quantity	star 1	star 2
mass [Msun]	36.78	6.709
phase	remnant	remnant
remnant	BH 36.78 Msun (delayed)	

remnant (2)			BH 6.709 Msun
separation		21.3 Rsun	
period		1.727 d	
eccentricity		0.2641	

4. Derived quantities

quantity	value	unit	formula
chirp mass M_c	12.81	Msun	$M_c = (m_1 m_2)^{3/5} / (m_1 + m_2)^{1/5}$
mass ratio q	0.1824	-	$q = m_2 / m_1$ (≤ 1)
symmetric mass ratio η	0.1305	-	$\eta = m_1 m_2 / (m_1 + m_2)^2$
integrated wind loss (star1 / star2)	1.132	Msun	$\int \dot{M}_{\text{wind},1} dt$; star2 = 0.561 Msun
integrated RLOF transfer	0	Msun	$\int \dot{M}_{\text{dot RLOF}} dt$ (gross, donor side)
net Delta M_1	5.218	Msun	$M_1(\text{ZAMS}) - M_1(\text{final})$
net Delta M_2	-26.29	Msun	$M_2(\text{final}) - M_2(\text{ZAMS})$
orbital separation (final)	21.3	Rsun	state variable
orbital period (final)	1.727	day	$P = 2 \pi \sqrt{a^3 / G(m_1 + m_2)}$
orbital binding energy	-2.1972e+49	erg	$E = -G m_1 m_2 / (2a)$
orbital angular momentum	1.0064e+54	$\text{g cm}^2 / \text{s}$	$J = \mu \sqrt{G M a (1 - e^2)}$
GW merger time	2.2163e+09	yr	Peters 1964 eq. 5.14 (eccentric)
GW merger time	2.216	Gyr	as above
peak GW frequency (ISCO)	101.1	Hz	$f = c^3 / (6^{3/2} \pi G M)$
char. strain at ISCO ($D=100$ Mpc)	6.0595e-21	-	Maggiore 2007 eq. 4.44

5. Error budget

****Numerical error**** (from the convergence study):

- not run in this invocation (--convergence flag)

****Physical / model uncertainty**** dominates and is quantified in §6-7. The reduced single-star evolution model (fitting formulae, not a structure)

6. Free-parameter sensitivity (one-at-a-time)

not run (--sensitivity).

7. Theory comparison (competing prescriptions)

not run (--theory-compare).

8. Validation suite results

test	status	detail
kepler_closure	PASS	200 orbits: max dE/E =1.77e-11, max dL/L =8.76e-12, da/a =1.52e-11 (tol 1e-06)
lane_emden_analytic	PASS	n=0: max dtheta =5.72e-16, dx11 =8.88e-16; n=1: max dtheta =1.55e-14, dx11 =1.82e-14; n=5: max dtheta =5.44e-14
periastron_advance_1PN	PASS	1PN d(omega)/orbit measured=6.157175e-05 rad, predicted 6 pi GM/(c^2 a(1-e^2))=6.155112e-05, frac diff=0.034%
hulse_taylor_decay	PASS	dPb/dt(Peters)=-2.402102e-12 s/s vs GR prediction -2.402630e-12 (Weisberg & Huang 2016); residual 0.022%. Merger
energy_conservation_2PN	PASS	2PN conservative: secular dE/E over 120 orbits = 1.68e-08 (tol 1e-07); max dL/L =1.12e-05
eddington_limit_flag	PASS	120 Msun ZAMS Gamma Edd=0.406; 10x-luminosity state flags Gamma>1: True
solar_structure_eddington	PASS	Eddington model 1 Msun: P_c=1.256e+17 (ref 2.4e17), T_c=1.185e+07 (ref 1.57e7), rho_c=76.9 (ref 150), L=0.0

9. Assumptions & limitations

- ****1-D, orbit-averaged.**** No 3-D hydrodynamics; the orbit is treated by secular equations except in the dedicated Kepler-closure test.
- ****Reduced stellar evolution.**** Single-star tracks are fitting formulae (Hurley+ 2000-style), not Henyey structure integrations. Core masses, radii and lifetimes carry ~10-30% systematic uncertainty.
- ****No magnetic fields**** beyond the Tayler-Spruit AM-transport toggle.
- ****No rotational mixing feedback**** on the tracks (rotation only sets

- Omega_crit and tidal rates here).
- **Mass-transfer rate** uses a Ritter/Kolb-Ritter-style analytic nozzle with an efficiency prefactor (KNOWN_GAPS BIN-MDOT).
 - **GW waveform** is a 0PN Newtonian chirp -- qualitative, not for matched filtering.
 - **3.5PN dynamics** not implemented (KNOWN_GAPS PN-35).
 - Full list: KNOWN_GAPS.md.

11. Bibliography

See `REFERENCES.bib` for the full list with DOIs. Key prescriptions used in this run: Peters (1964); Mora & Will (2004); Eggleton (1983); Ritte